Synthetic Network Troubleshooting Guide

Page 1: Commissioning Checks

Before commissioning a controller on an industrial Ethernet network, confirm that the documented network configuration matches the intended installation.

- Confirm the controller IP address, subnet mask and gateway against the approved network plan.
- Verify that each device has a unique address and that the connected switch port is enabled.
- Check the physical link indicators and inspect the Ethernet cable and connector condition.
- Record the tested configuration and the controller identifier before handover.

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Synthetic Network Troubleshooting Guide

Page 2: Communication Timeout

When communication times out, troubleshoot from the physical layer toward configuration rather than changing several settings at once.

- Check that the controller and network switch show an active physical link.
- Inspect the Ethernet cable, connector seating and switch port for an obvious interruption.
- Confirm the controller IP settings and the peer device address against the documented configuration.
- Review recent network or configuration changes and retest one change at a time.
- If the fault remains unresolved, capture the observed status and escalate with the relevant network details.

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Synthetic Network Troubleshooting Guide

Page 3: Duplicate IP and Intermittent Link Symptoms

Address conflicts can create intermittent communication that appears to recover and fail repeatedly.

- Symptoms can include alternating reachability, unexpected device identity, or communication that drops when another device starts.
- Compare configured device addresses with the network inventory and confirm that no duplicate IP address is present.
- If link indicators repeatedly drop, inspect cable routing, connector condition and switch-port logs before changing controller logic.
- Document the confirmed cause and update the network inventory after correction.

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